

Exploring the Sensitivity of Coulomb Stress Changes to Parameters Again through Strong Earthquakes in the Alaska Region

Lu Zhen

Abstract: The Coulomb stress change is useful to determine whether the fault will slip or not. We firstly review the derivation of error propagation applied in Coulomb stress change. Then we find related Alaska data and perform a concrete analysis. We find a similar result compared to previous studies. The difference between different models is obvious so that the first parameter should be considered separately from the other five parameters. Finally, we calculate the proportions of the remaining five parameters.

Key words: Coulomb stress change, error propagation, slip model

1 Background knowledge

Coulomb stress changes are widely used to analyze earthquake triggering. To calculate coulomb stress changes, we need source fault geometries, receiver fault geometries, and friction coefficient. For the robust determination of the Coulomb stress change, the sensitivity of the Coulomb stress change to these model parameters should be thoroughly assessed.

The Coulomb failure criterion is the basis of the Coulomb failure model: a fault is prone to sliding if the shear stress is greater than the static friction. Basic equation to calculate Coulomb stress change is shown as equation (1.1). If the coulomb stress change ΔCFF is greater than 0, the coulomb stress change accumulates. Parameters on the right side of equation (1.1) is not important, because they will be replaced by five more specific parts in following sections.

$$\Delta CFF = \Delta\tau + \mu(\Delta\sigma_n - B\frac{\Delta\sigma_{kk}}{3}) \quad (1.1)$$

2 Error propagation

Derivation consists of specification and linearization. (Figure 1) Firstly, we specify equation (1.1) and get equation (2.1). Equation (2.2) is Linearization form of equation (2.1).

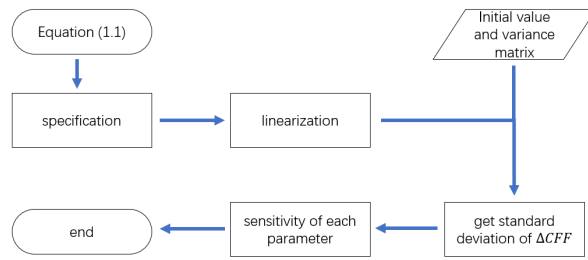


Figure 1: Flow chart

$$\begin{aligned} \Delta CFF = & \sin\lambda[-\frac{1}{2}\sin^2\phi\sin(2\delta)\sigma^{11} + \frac{1}{2}\sin(2\phi)\sin(2\delta)\sigma^{12} + \sin\phi\cos(2\delta)\sigma^{13} - \frac{1}{2}\cos^2\phi\sin(2\delta)\sigma^{22} \\ & - \cos\phi\cos(2\delta)\sigma^{23} + \frac{1}{2}\sin(2\delta)\sigma^{33}] + \cos\lambda[-\frac{1}{2}\sin(2\phi)\sin\delta\sigma^{11} + \cos(2\phi)\sin\delta\sigma^{12} \\ & + \cos\phi\cos\delta\sigma^{13} + \frac{1}{2}\sin(2\phi)\sin\delta\sigma^{22} + \sin\phi\cos\delta\sigma^{23}] + \mu[\sin^2\phi\sin^2\delta\sigma^{11} - \sin(2\phi)\sin^2\delta\sigma^{12} \\ & - \sin\phi\sin(2\delta)\sigma^{13} + \cos^2\phi\sin^2\delta\sigma^{22} + \cos\phi\sin(2\delta)\sigma^{23} + \cos^2\delta\sigma^{33} - \frac{B}{3}(\sigma^{11} + \sigma^{22} + \sigma^{33})] \end{aligned} \quad (2.1)$$

$$d\Delta CFF = \frac{\partial \Delta CFF}{\partial \sigma^{11}} d\sigma^{11} + \frac{\partial \Delta CFF}{\partial \sigma^{12}} d\sigma^{12} + \frac{\partial \Delta CFF}{\partial \sigma^{13}} d\sigma^{13} + \frac{\partial \Delta CFF}{\partial \sigma^{22}} d\sigma^{22} + \frac{\partial \Delta CFF}{\partial \sigma^{23}} d\sigma^{23} + \frac{\partial \Delta CFF}{\partial \sigma^{33}} d\sigma^{33} + \frac{\partial \Delta CFF}{\partial \phi} d\phi + \frac{\partial \Delta CFF}{\partial \delta} d\delta + \frac{\partial \Delta CFF}{\partial \lambda} d\lambda + \frac{\partial \Delta CFF}{\partial \mu} d\mu + \frac{\partial \Delta CFF}{\partial B} dB \quad (2.2)$$

In equation (2.2), there are totally 11 parameters on the right side. Let's set:

$$\frac{\partial \Delta CFF}{\partial \sigma^{11}} = k_{\sigma^{11}}, \frac{\partial \Delta CFF}{\partial \sigma^{12}} = k_{\sigma^{12}}, \frac{\partial \Delta CFF}{\partial \sigma^{13}} = k_{\sigma^{13}}, \frac{\partial \Delta CFF}{\partial \sigma^{22}} = k_{\sigma^{22}}, \frac{\partial \Delta CFF}{\partial \sigma^{23}} = k_{\sigma^{23}}, \frac{\partial \Delta CFF}{\partial \sigma^{33}} = k_{\sigma^{33}},$$

$$\frac{\partial \Delta CFF}{\partial \phi} = k_{\phi}, \frac{\partial \Delta CFF}{\partial \delta} = k_{\delta}, \frac{\partial \Delta CFF}{\partial \lambda} = k_{\lambda}, \frac{\partial \Delta CFF}{\partial \mu} = k_{\mu}, \frac{\partial \Delta CFF}{\partial B} = k_B.$$

Then we get equation (2.3).

$$d\Delta CFF = \begin{pmatrix} k_{\sigma^{11}} & k_{\sigma^{12}} & k_{\sigma^{13}} & k_{\sigma^{22}} & k_{\sigma^{23}} & k_{\sigma^{33}} & k_{\phi} & k_{\delta} & k_{\lambda} & k_{\mu} & k_B \end{pmatrix} \begin{pmatrix} d\sigma^{11} \\ d\sigma^{12} \\ d\sigma^{13} \\ d\sigma^{22} \\ d\sigma^{23} \\ d\sigma^{33} \\ d\phi \\ d\delta \\ d\lambda \\ d\mu \\ dB \end{pmatrix} \quad (2.3)$$

We define:

$$K = \begin{pmatrix} k_{\sigma^{11}} & k_{\sigma^{12}} & k_{\sigma^{13}} & k_{\sigma^{22}} & k_{\sigma^{23}} & k_{\sigma^{33}} & k_{\phi} & k_{\delta} & k_{\lambda} & k_{\mu} & k_B \end{pmatrix}$$

According to error propagation law, variance of coulomb stress change can be expressed as:

$$m_{\Delta CFF}^2 = K \begin{pmatrix} m_{\sigma^{11}}^2 & c_{\sigma^{11}}^{\sigma^{12}} & c_{\sigma^{11}}^{\sigma^{13}} & c_{\sigma^{11}}^{\sigma^{22}} & c_{\sigma^{11}}^{\sigma^{23}} & c_{\sigma^{11}}^{\sigma^{33}} & & & & & \\ c_{\sigma^{12}}^{\sigma^{11}} & m_{\sigma^{12}}^2 & c_{\sigma^{12}}^{\sigma^{13}} & c_{\sigma^{12}}^{\sigma^{22}} & c_{\sigma^{12}}^{\sigma^{23}} & c_{\sigma^{12}}^{\sigma^{33}} & & & & & \\ c_{\sigma^{13}}^{\sigma^{11}} & c_{\sigma^{13}}^{\sigma^{12}} & m_{\sigma^{13}}^2 & c_{\sigma^{13}}^{\sigma^{22}} & c_{\sigma^{13}}^{\sigma^{23}} & c_{\sigma^{13}}^{\sigma^{33}} & & & & & \\ c_{\sigma^{22}}^{\sigma^{11}} & c_{\sigma^{22}}^{\sigma^{12}} & c_{\sigma^{22}}^{\sigma^{13}} & m_{\sigma^{22}}^2 & c_{\sigma^{22}}^{\sigma^{23}} & c_{\sigma^{22}}^{\sigma^{33}} & & & & & \\ c_{\sigma^{23}}^{\sigma^{11}} & c_{\sigma^{23}}^{\sigma^{12}} & c_{\sigma^{23}}^{\sigma^{13}} & c_{\sigma^{23}}^{\sigma^{22}} & m_{\sigma^{23}}^2 & c_{\sigma^{23}}^{\sigma^{33}} & & & & & \\ c_{\sigma^{33}}^{\sigma^{11}} & c_{\sigma^{33}}^{\sigma^{12}} & c_{\sigma^{33}}^{\sigma^{13}} & c_{\sigma^{33}}^{\sigma^{22}} & c_{\sigma^{33}}^{\sigma^{23}} & m_{\sigma^{33}}^2 & & & & & \\ & & & & & & m_{\phi}^2 & & & & \\ & & & & & & & m_{\delta}^2 & & & \\ & & & & & & & & m_{\lambda}^2 & & \\ & & & & & & & & & m_{\mu}^2 & \\ & & & & & & & & & & m_B^2 \end{pmatrix} K^T \quad (2.4)$$

$c_{\sigma^{ij}}^{\sigma^{lm}}$ means covariance of σ^{ij} and σ^{lm} . The specific forms of coefficients are very complex and will not be displayed here. What matters is that the first 6 coefficients are only related to the last five parameters, and the last 5 coefficients have something to do with all parameters.

3 Analysis in the study region

Study region is located in interior Alaska. To analyze sensitivity of coulomb stress change to different parameters, we need know coefficient vector and variance-covariance matrix in equation (2.4).

Firstly, we get 5 different focal mechanisms. Then we totally find 6 slip models of our target earthquake. Based on these data and some empirical results, we can complete the calculation.

After calculating, we find that the right side of equation (2.4) can be divided into 2 parts of equal magnitude. The first part includes the first 6 parameters, and the second part consists of the last 5 parameters.

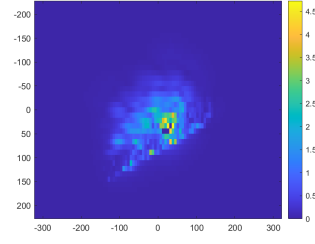


Figure 2: variance of the first part(Bar for colorbar, km for axes)

Because of features mentioned at the end of section 1, the variance contributed by first part in 6 different models are equal, which is indicated in Figure 2. However, variance contributed by the second part differs from each other, which is indicated in Figure 3.

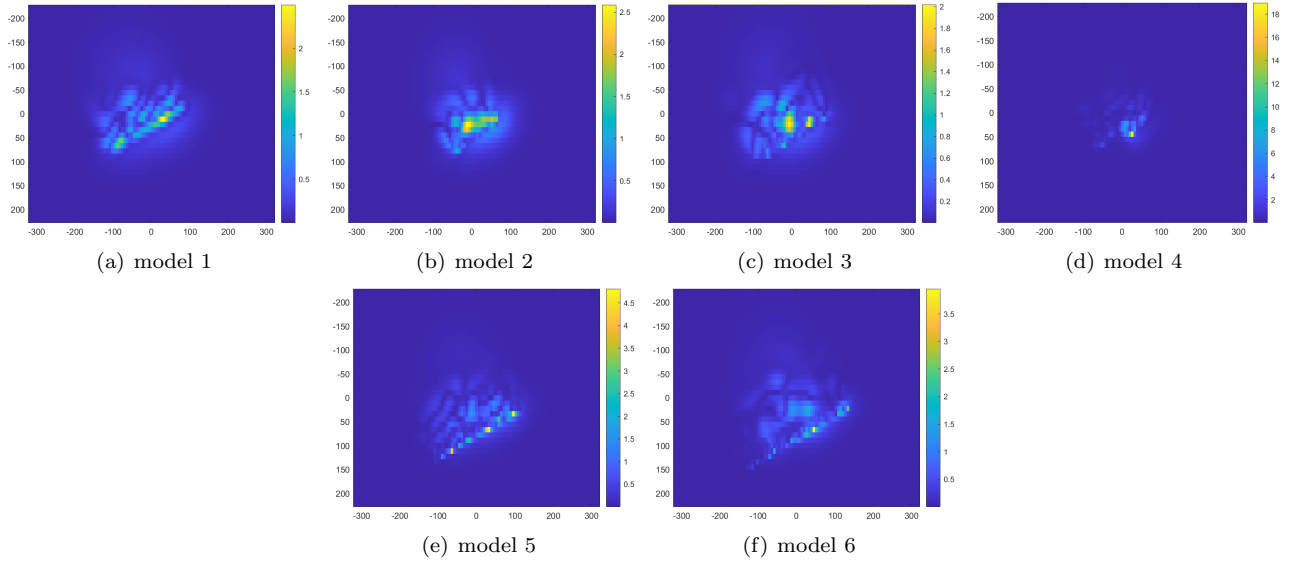


Figure 3: variance of the second part(Bar for colorbar, km for axes)

Additionally, we calculate the average proportion of different parameters in the second part, which is displayed in Table 1. Apparently, except of the only empirical parameter, B , ϕ plays the most important role.

Table 1: proportions of last 5 parameters

model	$ k_\phi /m_\phi$	$ k_\delta /m_\delta$	$ k_\lambda /m_\lambda$	$ k_\mu /m_\mu$	$ k_B /m_B$
1	0.3358	0.0711	0.1672	0.0489	0.3769
2	0.3384	0.0715	0.1450	0.0464	0.3987
3	0.3429	0.0780	0.1568	0.0509	0.3713
4	0.3389	0.0723	0.1353	0.0511	0.4024
5	0.3554	0.0711	0.1542	0.0464	0.3729
6	0.3432	0.0732	0.1602	0.0458	0.3777